SMART BATTERY MANAGEMENT SYSTEM

Battery Condition Technical Report

REPORT ID: BMS-TEST-001
GENERATED: 2026-10-20 13:22:10 UTC

REPORTING PERIOD: 24H
BATTERY BUS: 12.0V Nominal

SPECIFICATION STATUS: UNVERIFIED PROTOTYPE · Profile: **Nominal 12V Li-ion Prototype** (Unverified). Exact chemistry and model are unverified. Provisional safety margins are applied. Do not exceed 45°C ambient or provisional voltage bounds.

SECTION A: OVERALL OBSERVED CONDITION

STATUS: NORMAL

Nominal 12V prototype operating within expected bounds.

SECTION B: LATEST VALID SENSOR MEASUREMENTS

Directly sampled from hardware registers via Firebase Realtime Database (/smart_bms/latest).

Parameter	Measured Value	Hardware / Path	Data Status	Technical Context / Scope
Bus Voltage	**12.35 V**	INA219 (I²C 0x40)	**VALID**	Battery terminal potential across high-side shunt.
Shunt Current	**0.850 A**	INA219 (I²C 0x40)	**VALID**	Positive: Charge / Negative: Discharge across 0.1Ω.
Active Power	**10.50 W**	Calculated (V × I)	VALID	Real-time electrical power transferred to load bus.
Ambient Temperature	**28.4 °C**	DHT11 (1-Wire D7)	**VALID**	**Enclosure ambient air** near pack (NOT cell core).
Ambient Humidity	**55.0 %**	DHT11 (1-Wire D7)	VALID	Relative air humidity inside prototype enclosure.

SECTION C: KEY FINDINGS & OBSERVATIONS

1. Monitored nominal 12V bus voltage recorded at 12.35 V.
2. Ambient enclosure temperature measured at 28.4 °C; cooling fan threshold at 35.0°C.
3. No immediate hardware safety trip conditions detected in latest sample.